

Winning Space Race with Data Science

Joseph Appolos

4 November 2026

GitHub repository github.com/SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction

Outline

1. **Executive Summary** Methods and headline results
2. **Introduction** Background, context and questions
3. **Methodology** Collection, wrangling, EDA, interactive analytics, ML
4. **Results** EDA, SQL, Folium, Dash, classification
5. **Innovative Insights** Experience curve, hardware, bid implications
6. **Conclusion** What the data says
7. **Appendix** Repository, notebooks and code



The four SpaceX launch sites analysed (Folium)

Executive Summary

SUMMARY OF METHODOLOGIES

- Collected launch data from the SpaceX REST API and by web-scraping Wikipedia's Falcon 9 launch tables
- Wrangled the data and derived a binary landing label (Class) from 8 outcome types
- Explored the data with SQL (SQLite) and Matplotlib/Seaborn charts
- Built interactive analytics: Folium launch-site maps and a Plotly Dash dashboard
- Trained and tuned Logistic Regression, SVM, Decision Tree and KNN with 10-fold GridSearchCV

SUMMARY OF RESULTS

- **66.7%** of 90 Falcon 9 first stages landed; success rose from 43% (flights 1–30) to 83% (flights 61–90)
- **KSC LC-39A** has the highest launch success rate (76.9%); ES-L1, GEO, HEO and SSO orbits landed 100%
- **Sites** are 1–2 km from coast, railway and highway, but ~15 km from the nearest city
- **83.3%** test accuracy for all four models; Decision Tree best in cross-validation (86.1%)
- **Zero missed landings,** but 3 of 6 failures were predicted as landings (false positives)

KEY INSIGHT We can predict first-stage landings with about 83% accuracy, which helps estimate the cost of a SpaceX launch.

Introduction

PROJECT BACKGROUND AND CONTEXT

- SpaceX advertises Falcon 9 launches at $62M; other providers charge $165M+
- Most of the saving comes from recovering and reusing the first stage
- If we can predict whether the first stage lands, we can estimate the real cost of a launch
- As data scientists for a new competitor (Space Y), we use public launch data to inform competitive bids

PROBLEMS WE WANT TO ANSWER

- Which factors (payload, orbit, site, booster, experience) drive landing success?
- How do launch-site locations relate to safety and logistics?
- Can a classifier predict a landing accurately enough to support pricing?

$62M
Falcon 9 list price

$165M+
Typical competitor price

90
Falcon 9 launches modelled

4
Launch sites analysed

About 62% cheaper, mostly because SpaceX reuses the first stage.

KEY INSIGHT If we can predict whether the first stage lands, we can estimate the cost of the launch.

Section 1

Methodology

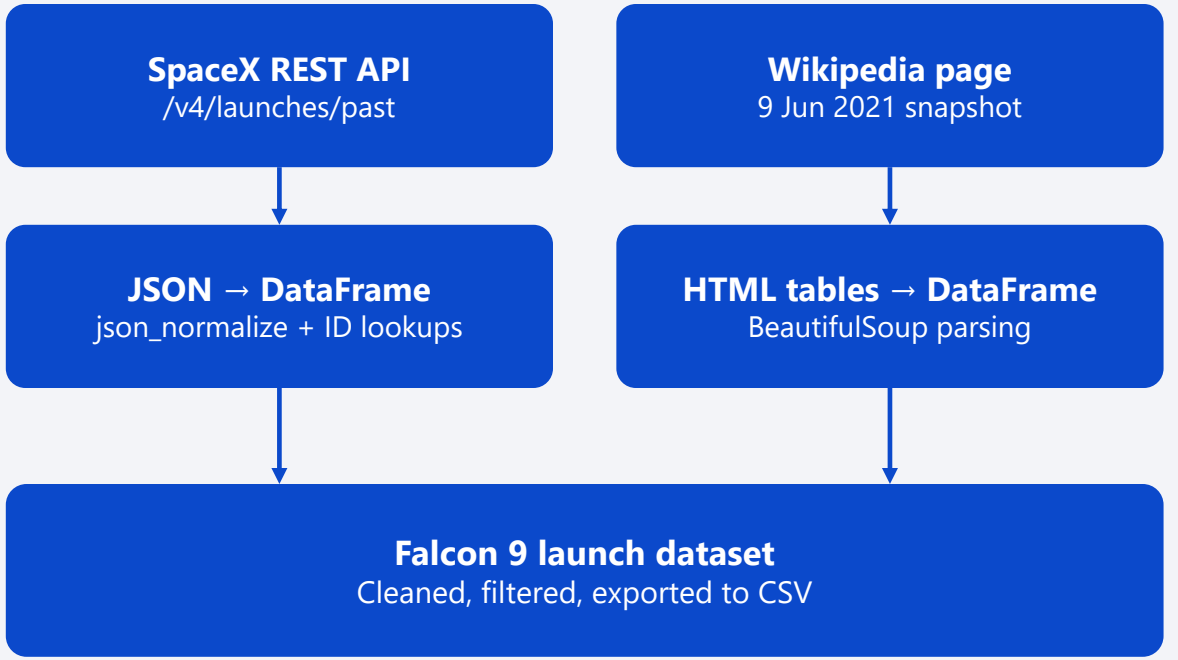


Methodology

1. **Data collection:** SpaceX REST API (static JSON snapshot) + BeautifulSoup scraping of Wikipedia launch tables
2. **Data wrangling:** Filtered to Falcon 9, imputed payload mass, mapped 8 landing outcomes to a 0/1 Class label
3. **EDA with visualization & SQL:** Seaborn scatter, bar and line charts; 10 SQL queries in SQLite
4. **Interactive visual analytics:** Folium maps of sites, outcomes and proximities; Plotly Dash dashboard
5. **Predictive analysis:** Standardized features, 80/20 split, GridSearchCV (cv=10) over 4 classifiers
6. **Evaluation:** Test accuracy and confusion matrices to choose the best model

Data Collection

- **Two complementary sources:** the SpaceX REST API (launch, rocket, launchpad, payload and core endpoints) and Wikipedia's "List of Falcon 9 and Falcon Heavy launches"
- **API:** GET request → JSON → pandas json_normalize; IDs resolved to booster name, site coordinates, payload mass, orbit and landing details
- **Web scraping:** requests + BeautifulSoup on the 9 June 2021 page snapshot; 121 launch rows parsed from the HTML tables
- **Output:** clean CSVs (dataset_part_1, spacex_web_scraped) feeding wrangling, EDA and modelling

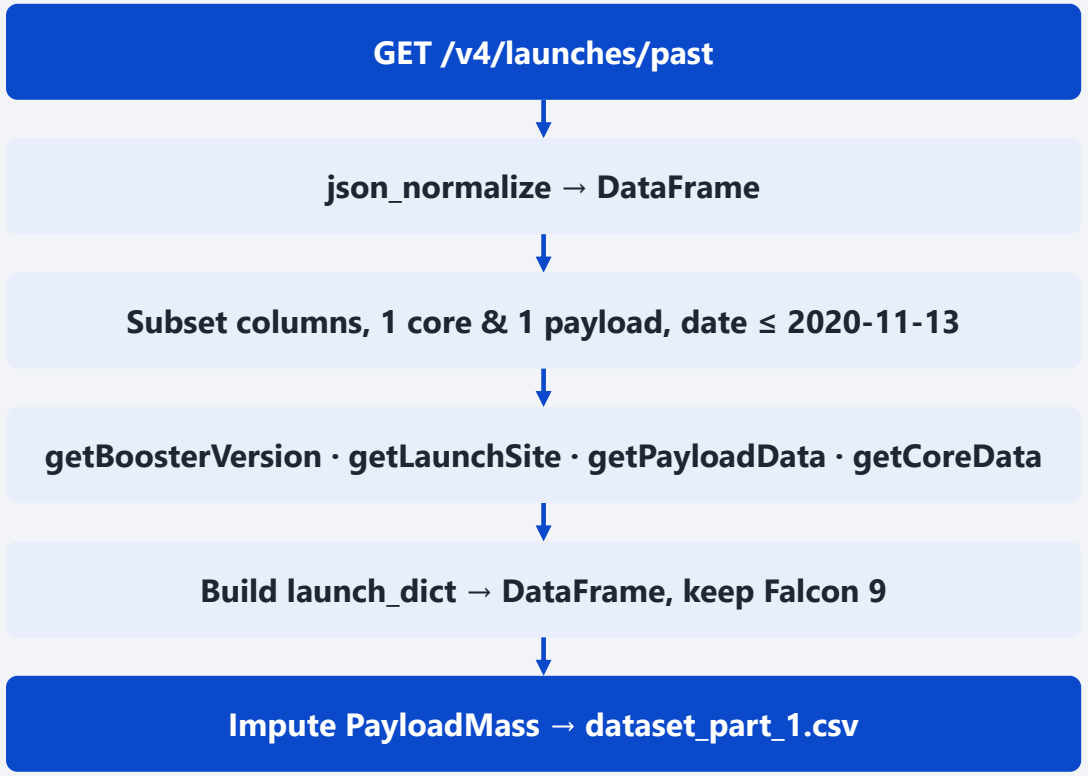


KEY INSIGHT Combining the API data with the Wikipedia tables gives a more complete launch history.

Data Collection – SpaceX API

- **Request:** requests.get() on the SpaceX v4 launches endpoint (static JSON copy provided in the lab)
- **Parse:** response.json() → pd.json_normalize()
- **Enrich:** helper functions call /rockets, /launchpads, /payloads and /cores by ID
- **Clean:** keep single-core, single-payload launches up to 13 Nov 2020; keep Falcon 9 only; fill missing PayloadMass with the mean

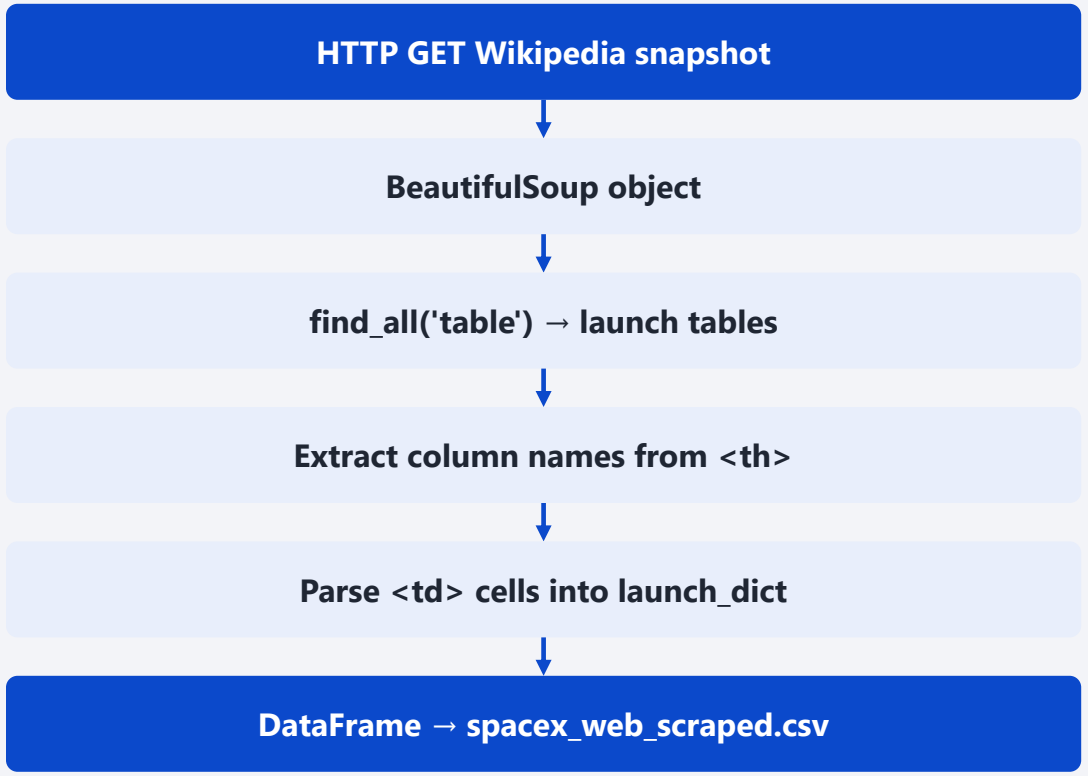
GitHub URL: https://github.com/SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction/blob/main/notebooks/01_data_collection_api.ipynb



Data Collection - Scraping

- **Request:** requests.get() on the Wikipedia snapshot URL with a browser User-Agent header
- **Parse:** BeautifulSoup(html, 'html.parser'); verified via soup.title
- **Columns:** find_all('th') + extract_column_from_header() on the first launch table
- **Rows:** loop every 'wikitable plainrowheaders collapsible' table; keep rows whose header is a flight number
- **Fields:** flight no., date, time, booster version, site, payload, mass, orbit, customer, launch outcome, booster landing
- **Result:** 121 launch records → DataFrame → spacex_web_scraped.csv

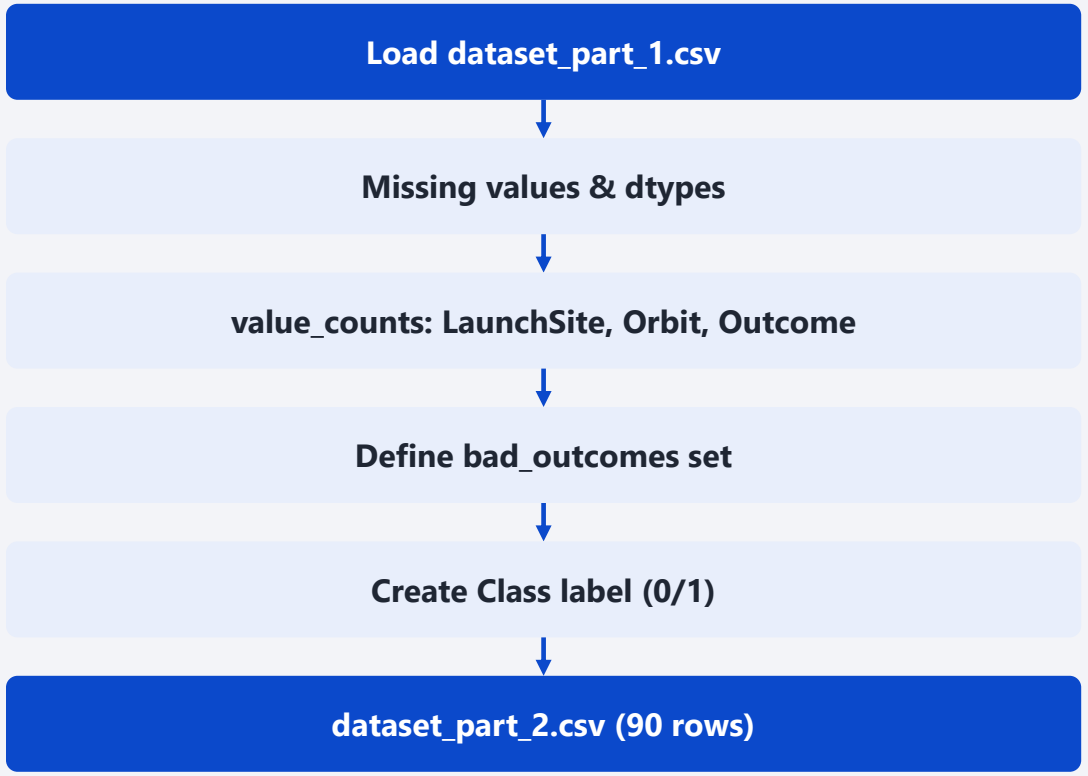
GitHub URL: https://github.com/SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction/blob/main/notebooks/02_data_collection_webscraping.ipynb



Data Wrangling

- **Profile:** % missing per column (only LandingPad, 28.9%) and numeric vs categorical dtypes
- **Count:** launches per site (CCAFS SLC 40: 55, KSC LC 39A: 22, VAFB SLC 4E: 13) and per orbit
- **Outcomes:** 8 landing outcomes, e.g. True ASDS (41), None None (19), True RTLS (14)
- **Label:** outcomes in bad_outcomes → 0, all others → 1 (Class)
- **Result:** 60 landed / 30 not landed → 66.7% success; saved as dataset_part_2.csv

Outcome	Meaning	Class
True ASDS / RTLS / Ocean	Landed on drone ship / ground pad / ocean	1
False ASDS / RTLS / Ocean	Landing attempted, failed	0
None ASDS / None None	No landing / not attempted	0



GitHub URL: https://github.com/SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction/blob/main/notebooks/03_data_wrangling.ipynb

EDA with Data Visualization

Chart	Type	Why we used it
Flight Number vs. Payload Mass	Scatter (catplot, hue = Class)	See whether experience and payload jointly affect landing
Flight Number vs. Launch Site	Scatter (catplot)	Compare how each site's record evolves over time
Payload Mass vs. Launch Site	Scatter (catplot)	Check which sites handle heavy payloads and how they fare
Success rate by Orbit type	Bar chart (groupby mean)	Compare a rate across categories
Flight Number vs. Orbit type	Scatter (catplot)	Detect orbit-specific learning effects
Payload Mass vs. Orbit type	Scatter (catplot)	Detect payload limits / risk by orbit
Yearly success rate	Line chart	Show the trend over time (2010–2020)

Feature engineering: selected 12 features and one-hot encoded Orbit, LaunchSite, LandingPad and Serial with get_dummies(), then cast all to float64 → dataset_part_3.csv for modelling

GitHub URL: https://github.com/SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction/blob/main/notebooks/05_eda_visualization.ipynb

EDA with SQL

1. **Unique launch sites**
   `SELECT DISTINCT Launch_Site`

2. **5 sites beginning with 'CCA'**
   `LIKE 'CCA%' … LIMIT 5`

3. **Total payload for NASA (CRS)**
   `SUM(PAYLOAD_MASS__KG_)`

4. **Average payload of F9 v1.1**
   `AVG(…) with LIKE '%F9 v1.1%'`

5. **First successful ground-pad landing**
   `filter + ORDER BY Date LIMIT 1`

6. **Drone-ship successes, 4000–6000 kg**
   `WHERE … BETWEEN 4000 AND 6000`

7. **Mission outcome totals**
   `GROUP BY Mission_Outcome + COUNT`

8. **Boosters with max payload**
   `subquery with MAX()`

9. **2015 drone-ship failures by month**
   `substr(Date,6,2), substr(Date,1,4)='2015'`

10. **Rank landing outcomes 2010-06-04 → 2017-03-20**
    `GROUP BY … ORDER BY count DESC`

Data loaded from Spacex.csv into SQLite (SPACEXTBL → SPACEXTABLE, blank rows removed) and queried with the %%sql magic.

GitHub URL: https://github.com/SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction/blob/main/notebooks/04_eda_sql.ipynb

Build an Interactive Map with Folium

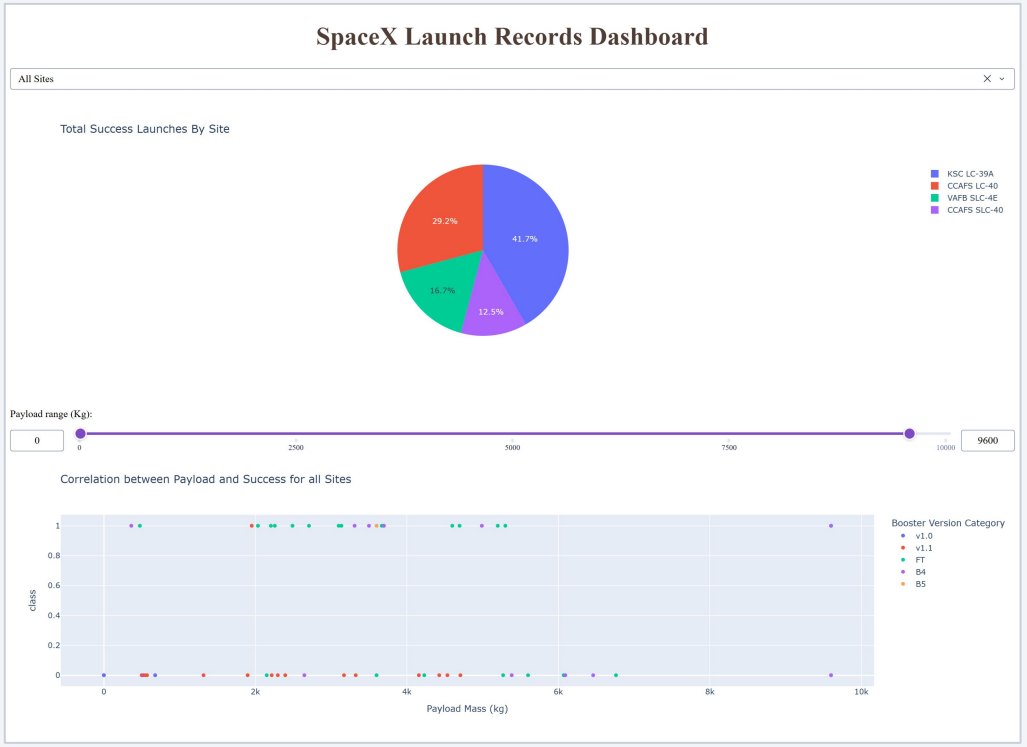
Map object	What we added	Why
folium.Circle + Popup	1 km circle at each of the 4 launch sites	Highlight the site area and name on click
Marker + DivIcon	Text label with each site name	Identify sites without clicking
MarkerCluster	One marker per launch, grouped by site	56 launches share 4 coordinates; clusters keep the map readable
Marker icon colour	Green = landed (class 1), red = failed (class 0)	See each site's success rate at a glance
MousePosition	Live Lat/Long readout	Pick coordinates of coastline, rail, highway and city
PolyLine + distance DivIcon	Lines from VAFB SLC-4E to proximities, labelled in km (haversine)	Quantify how close sites are to infrastructure and cities

GitHub URL: https://github.com/SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction/blob/main/notebooks/06_interactive_map_folium.ipynb

KEY INSIGHT Showing sites, outcomes and distances on one map makes the location patterns easy to see.

Build a Dashboard with Plotly Dash

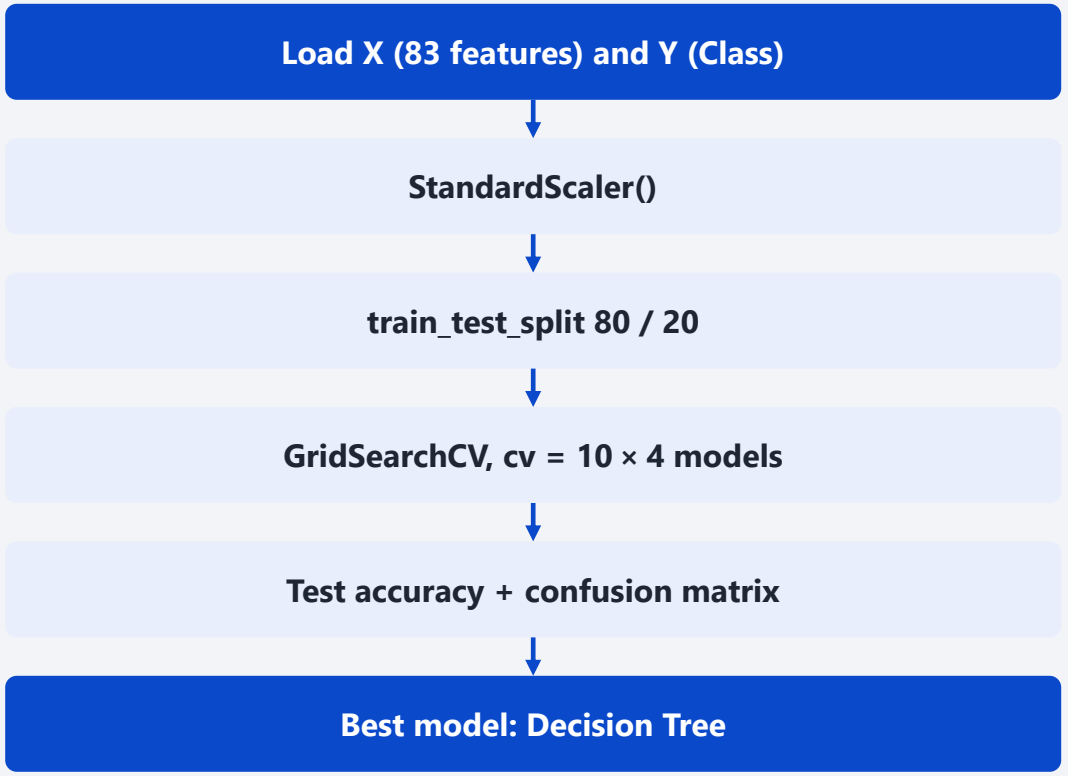
- **Launch-site dropdown:** All Sites or one of 4 sites; drives both charts
- **Success pie chart:** All Sites → share of successful launches per site; one site → success vs. failure split
- **Payload range slider:** 0–10,000 kg in 1,000 kg steps to filter launches by payload
- **Payload vs. outcome scatter:** class (0/1) against payload mass, coloured by booster version category
- **Callbacks:** two @app.callback functions recompute the figures whenever the dropdown or slider changes
- **Why:** lets stakeholders answer "which site / payload range / booster works best?" themselves, without code



GitHub URL: https://github.com/SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction/blob/main/dashboard/spacex_dash_app.py

Predictive Analysis (Classification)

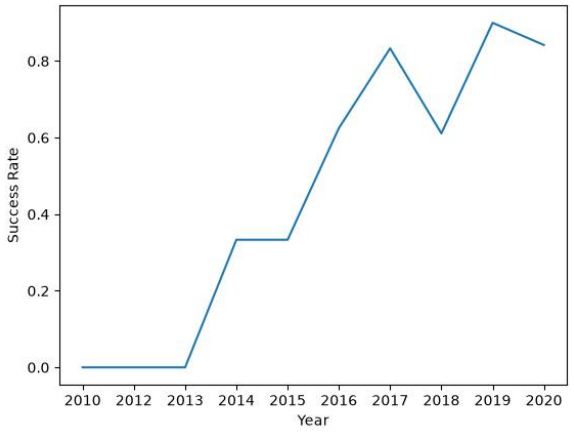
- **Data:** Y = Class from dataset_part_2; X = 83 one-hot features from dataset_part_3
- **Pre-process:** StandardScaler; train_test_split(test_size=0.2, random_state=2) → 72 train / 18 test
- **Build & tune:** GridSearchCV(cv=10) for Logistic Regression (C), SVM (kernel, C, gamma), Decision Tree (criterion, splitter, depth, leaf/split sizes) and KNN (k, algorithm, p)
- **Evaluate:** best_score_ (CV accuracy), .score() on the test set and a confusion matrix per model
- **Select:** the model with the highest CV accuracy, checked against test accuracy



GitHub URL: https://github.com/SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction/blob/main/notebooks/07_machine_learning_prediction.ipynb

Results

Exploratory data analysis results



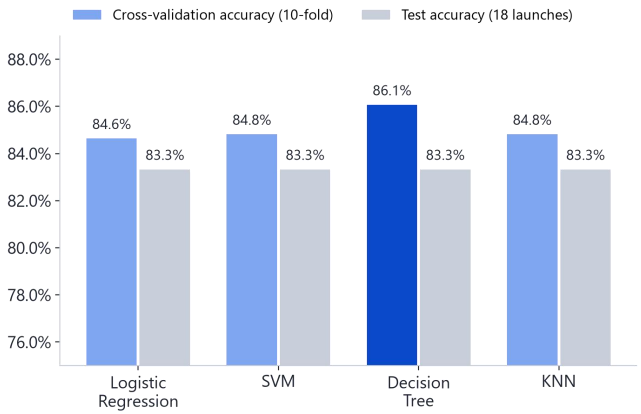
Success climbs from 0% (2010–13) to 84–90% (2019–20)

Interactive analytics demo in screenshots



Folium maps + Dash dashboard of sites and payloads

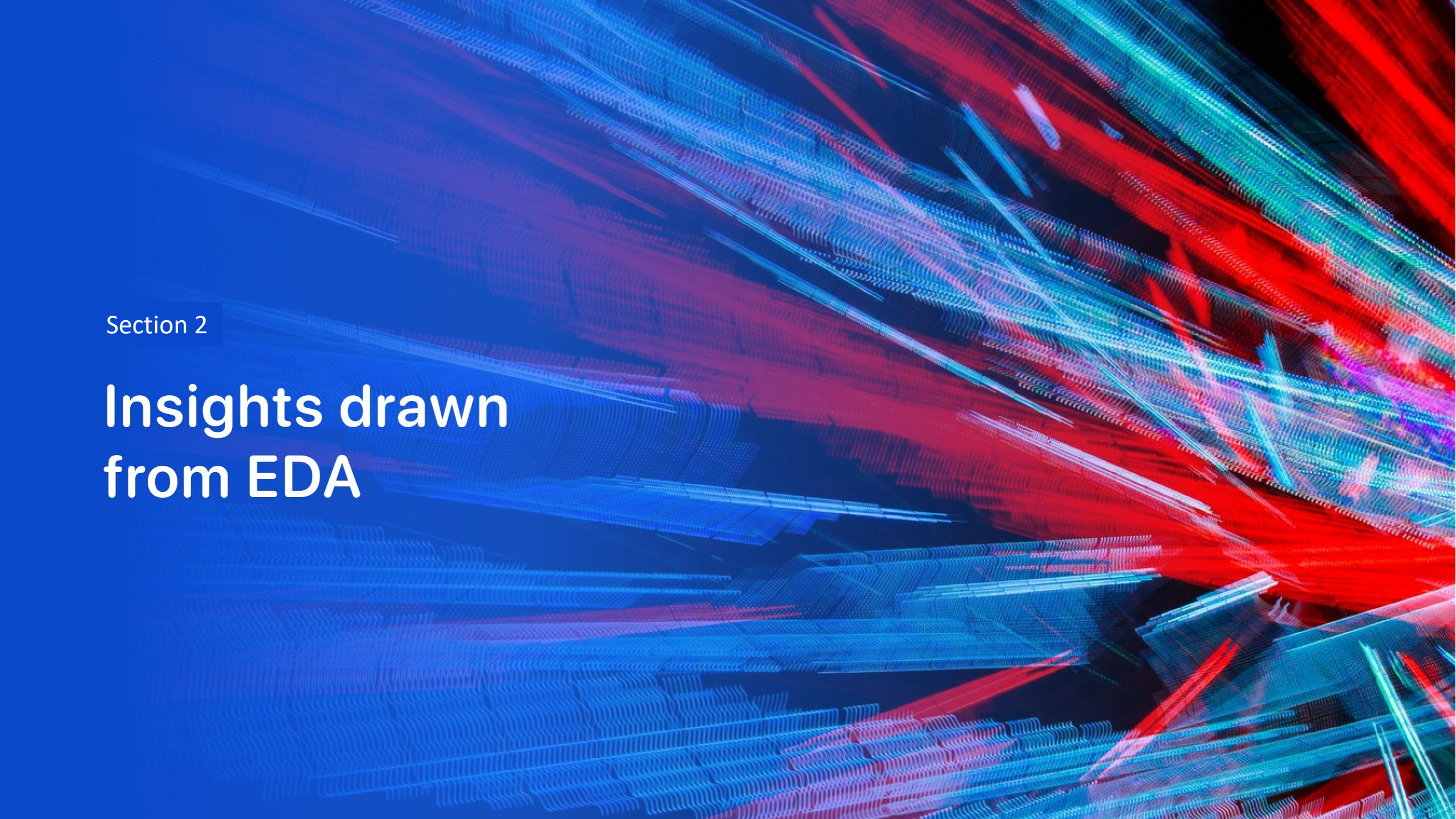
Predictive analysis results



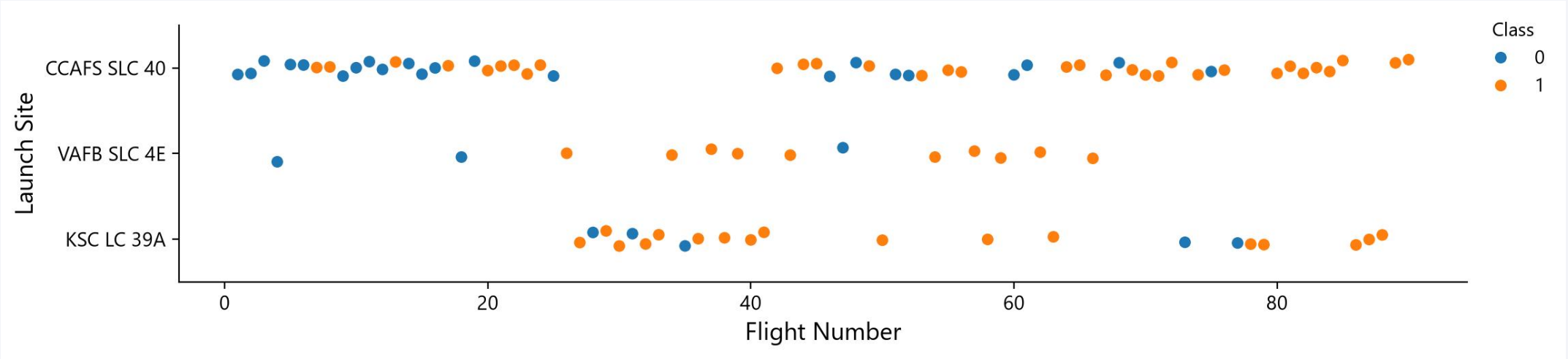
4 tuned classifiers, 83.3% test accuracy

Section 2

Insights drawn from EDA



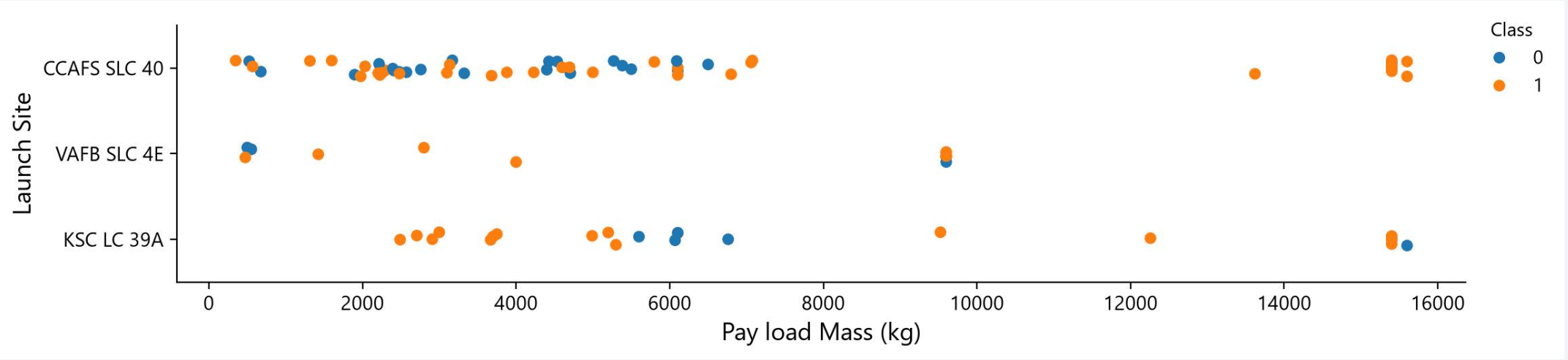
Flight Number vs. Launch Site



- CCAFS SLC 40 carried most early flights (first 20 flights there landed only 28% of the time)
- KSC LC 39A only started at flight 27, after the learning phase, and is mostly successful
- At every site, later flights (higher flight numbers) are predominantly successful

KEY INSIGHT Success improves with flight number at every launch site.

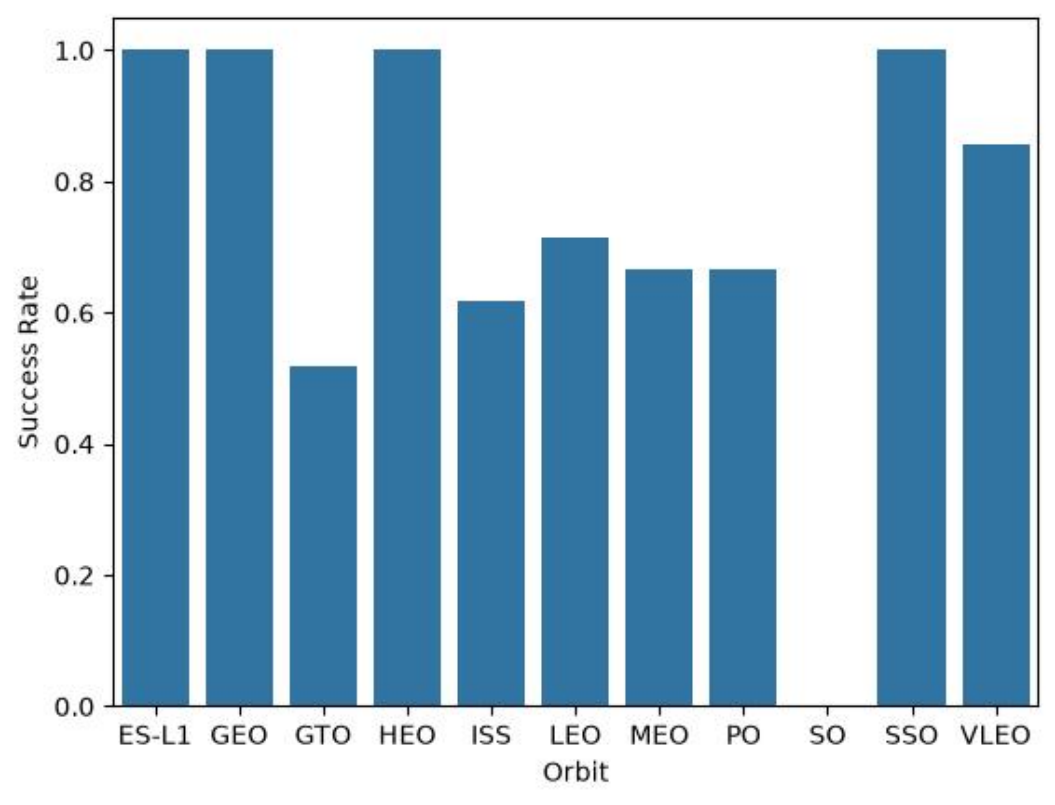
Payload vs. Launch Site



- VAFB SLC 4E never launched payloads above ~10,000 kg (max 9,600 kg)
- Heavy payloads (>10,000 kg) flew only from CCAFS SLC 40 (89% success) and KSC LC 39A (83%)
- Light payloads at CCAFS show a mix of outcomes, mostly from early flights

KEY INSIGHT Heavy payloads (over 10,000 kg) still landed well, mostly because they were later flights.

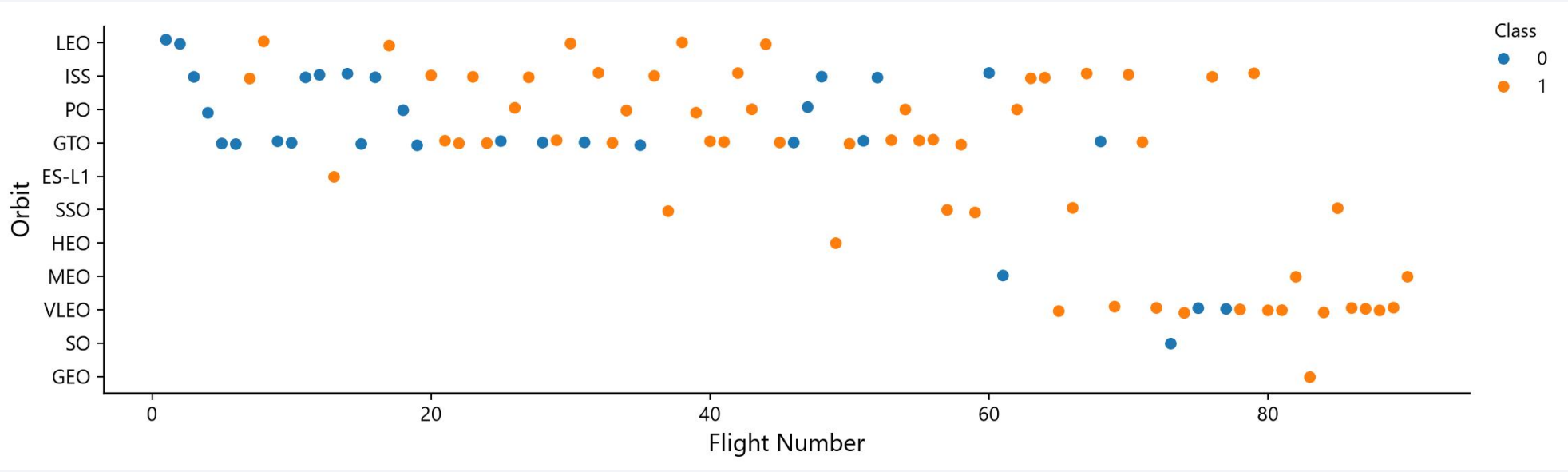
Success Rate vs. Orbit Type



- ES-L1, GEO, HEO and SSO: 100% success (but only 1, 1, 1 and 5 launches)
- VLEO 86% (14 launches), LEO 71%, PO / MEO 67%, ISS 62%
- GTO is the hardest high-volume orbit at 52% (27 launches); SO 0% (1 launch)

KEY INSIGHT GTO has the lowest success rate of the common orbits. The 100% orbits only have a few launches.

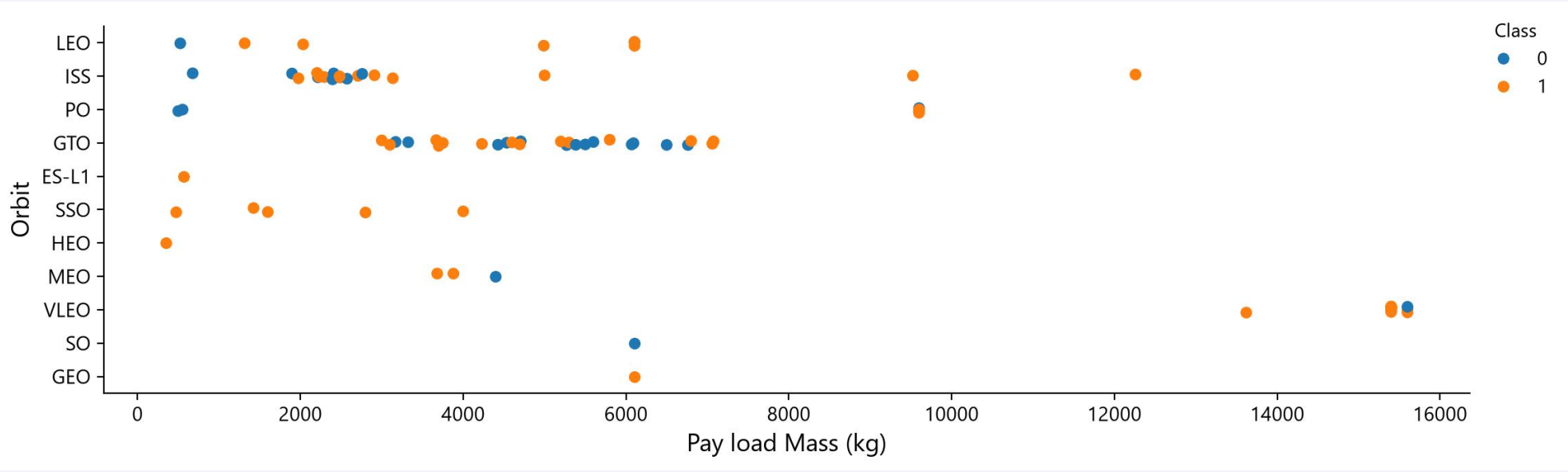
Flight Number vs. Orbit Type



- LEO: success appears related to flight number (both failures were flights 1–2)
- GTO: no clear relationship; failures persist across early and later flights
- VLEO launches only began at flight 65 and are almost all successful

KEY INSIGHT In LEO, success grows with the number of flights, while GTO stays mixed.

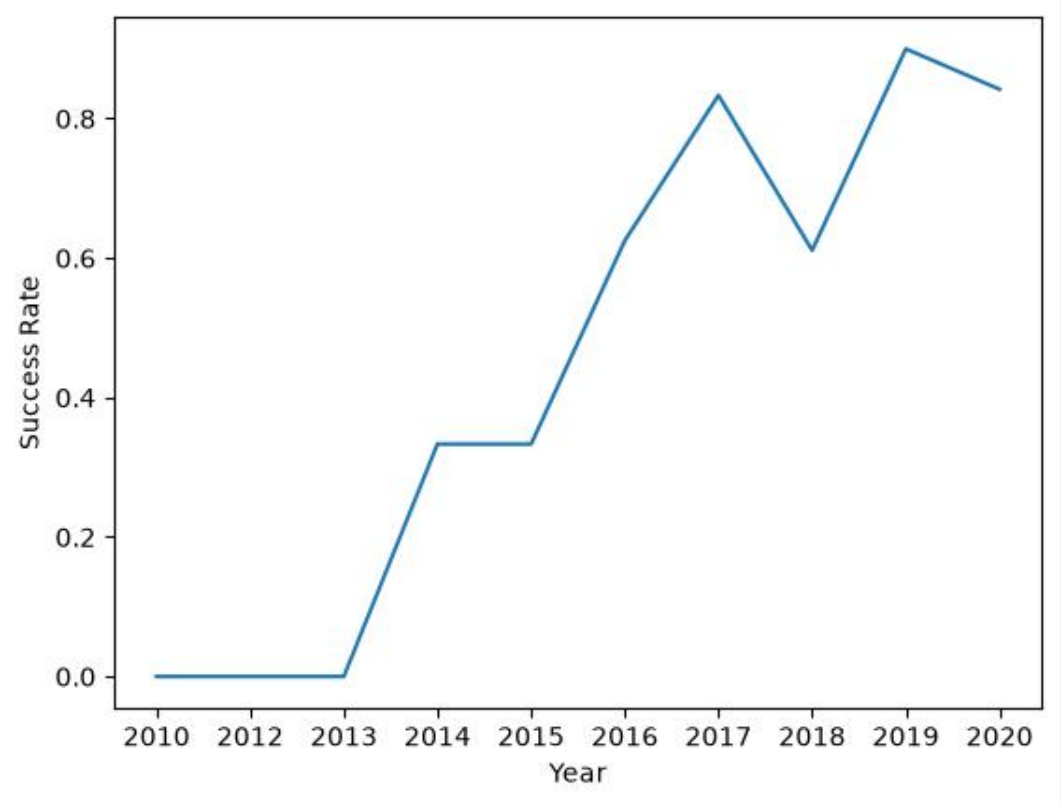
Payload vs. Orbit Type



- Heavy payloads land successfully more often for Polar (PO), LEO and ISS missions
- GTO payloads (3,000–7,000 kg) show both outcomes, so payload does not separate them
- The heaviest loads (~15,600 kg) are VLEO (Starlink) and mostly land

KEY INSIGHT The effect of payload mass depends on the orbit type.

Launch Success Yearly Trend



- 0% success from 2010 to 2013, while landing techniques were developed
- Steady climb from 2013 to 83% in 2017; dip to 61% in 2018 (Block 5 transition)
- 90% in 2019 and 84% in 2020

KEY INSIGHT The success rate kept increasing from 2013 to 2020.

All Launch Site Names

QUERY

```
%%sql
select distinct "Launch_Site" from
spacextable
```

EXPLANATION

- SELECT DISTINCT returns each launch site once
- 4 sites: 3 in Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and 1 in California (VAFB SLC-4E)
- CCAFS LC-40 and CCAFS SLC-40 are Launch Complex 40 under its older and newer names

RESULT

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

KEY INSIGHT All launches in the dataset come from these 4 sites.

Launch Site Names Begin with 'CCA'

QUERY

```
%%sql
select *
from spacextable
where "Launch_Site" like 'CCA%'
Limit 5
```

- LIKE 'CCA%' matches any site starting with CCA; LIMIT 5 returns the first 5 records
- All five are early Cape Canaveral (CCAFS LC-40) flights of F9 v1.0, 2010–2013, to LEO/ISS

Date	Booster	Site	Payload	kg	Orbit	Customer	Mission	Landing
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Un...	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSat...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

KEY INSIGHT None of these first five launches recovered the first stage.

Total Payload Mass

QUERY

```
%%sql
select sum("PAYLOAD_MASS__KG_") as "Total Payload Mass for
NASA CRS"
from spacextable
where Customer = "NASA (CRS)"
```

RESULT

Total Payload Mass for NASA CRS (kg)
45,596

EXPLANATION

- SUM() adds PAYLOAD_MASS__KG_ for launches whose customer is exactly NASA (CRS)
- Exact match excludes a shared ride (“NASA (CRS), Kacific 1”, 2,617 kg)
- 45,596 kg carried across 20 Commercial Resupply Services missions to the ISS

KEY INSIGHT NASA CRS missions carried a total of 45,596 kg.

Average Payload Mass by F9 v1.1

QUERY

```
%%sql
select avg("payload_mass__kg_") as "averge payload mass"
from spacextable
where "booster_version" like "%F9 v1.1%"
```

RESULT

Average payload mass (kg)
2,534.67

EXPLANATION

- AVG() over all boosters whose version contains “F9 v1.1”
- F9 v1.1 carried ~2,535 kg on average, well below later Full Thrust / Block 5 boosters
- This early version also had almost no successful landings (1 of 15 in the dashboard data)

KEY INSIGHT F9 v1.1 carried fairly light payloads, but it rarely landed.

First Successful Ground Landing Date

QUERY

```
%%sql
select Date
from spacextable
where "Landing_Outcome" = 'Success (ground pad)'
order by Date
limit 1
```

RESULT

First successful ground-pad landing
2015-12-22

EXPLANATION

- Filter Landing_Outcome = 'Success (ground pad)', sort by date, take the first row
- 22 December 2015: booster F9 FT B1019 (Orbcomm OG2 mission) landed at Cape Canaveral
- Came ~5.5 years after the first Falcon 9 flight (June 2010)

KEY INSIGHT The first successful ground pad landing was on 22 December 2015.

Drone Ship Landings, 4,000–6,000 kg Payload

QUERY

```
%%sql
select "Booster_Version"
from spacextable
where "payload_mass__kg_" between 4000 and 6000
and "landing_outcome"  = "Success (drone ship)"
```

RESULT

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

EXPLANATION

- Filters landing outcome Success (drone ship) and payload between 4,000 and 6,000 kg
- 4 boosters qualify, all F9 Full Thrust (FT)
- Two (B1021.2, B1031.2) were re-flown boosters, early proof of reuse

KEY INSIGHT All four boosters are F9 FT, and two of them were re-flown boosters.

Total Number of Successful and Failure Mission Outcomes

QUERY

```
%%sql
select "mission_outcome", count("mission_outcome") as "Count"
from spacextable
group by "mission_outcome"
```

EXPLANATION

- GROUP BY Mission_Outcome with COUNT()
- 100 successful missions vs. 1 failure (in flight)
- 'Success' appears twice because one value has a trailing space, a data-quality issue worth cleaning

RESULT

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1
Total success / failure	100 / 1

KEY INSIGHT Almost every mission reached orbit (100 of 101), so landing the first stage is the harder part.

Boosters Carried Maximum Payload

QUERY

```
%%sql
select "booster_version"
    from spacextable
    where "payload_mass__kg_" =  (select
max("payload_mass__kg_") from spacextable)
```

EXPLANATION

- Subquery finds MAX(PAYLOAD_MASS__KG_) = 15,600 kg; outer query lists boosters with that mass
- 12 launches, all Block 5 (F9 B5) boosters
- The same cores (B1048, B1049, B1051, B1060…) fly repeatedly at maximum load

RESULT

Booster	Booster	Booster
F9 B5 B1048.4	F9 B5 B1049.4	F9 B5 B1051.3
F9 B5 B1056.4	F9 B5 B1048.5	F9 B5 B1051.4
F9 B5 B1049.5	F9 B5 B1060.2	F9 B5 B1058.3
F9 B5 B1051.6	F9 B5 B1060.3	F9 B5 B1049.7

KEY INSIGHT Only Block 5 boosters carried the maximum payload of 15,600 kg.

2015 Launch Records

QUERY

```
%%sql
SELECT substr("Date", 6, 2) AS "Month", "Landing_Outcome",
"Booster_Version", "Launch_Site"
FROM SPACEXTABLE
WHERE substr("Date", 1, 4) = '2015'
AND "Landing_Outcome" LIKE 'Failure (drone ship)%'
```

RESULT

Month	Landing_Outcome	Booster	Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

EXPLANATION

- substr(Date,6,2) extracts the month; substr(Date,1,4) = '2015' filters the year (SQLite has no month names)
- Two drone-ship failures in 2015: January (F9 v1.1 B1012) and April (F9 v1.1 B1015), both at CCAFS LC-40

KEY INSIGHT Both 2015 drone ship failures were F9 v1.1 boosters launched from CCAFS LC-40.

Rank Landing Outcomes (2010-06-04 to 2017-03-20)

QUERY

```
%%sql
select "landing_outcome", count("landing_outcome") as
"outcomes"
from spacextable
where "Date" between "2010-06-04" and "2017-03-20"
group by "landing_outcome"
order by outcomes desc
```

EXPLANATION

- Counts each landing outcome in the window, grouped and ordered by count (descending)
- "No attempt" (10) leads: early missions often did not try to land
- Drone-ship successes and failures are tied at 5 each

RESULT

Rank	Landing_Outcome	Count
1	No attempt	10
2	Success (drone ship)	5
3	Failure (drone ship)	5
4	Success (ground pad)	3
5	Controlled (ocean)	3
6	Uncontrolled (ocean)	2
7	Failure (parachute)	2
8	Precluded (drone ship)	1

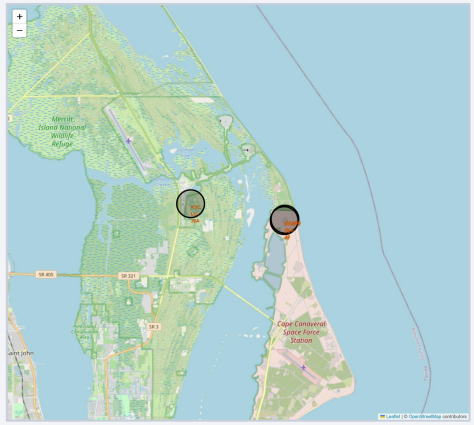
KEY INSIGHT Between 2010 and March 2017, only 8 of 31 outcomes were successful drone ship or ground pad landings.


Section 3
Launch Sites
Proximities Analysis

Launch Sites on the Global Map



All 4 sites: circles with name labels (Folium)



Cape Canaveral zoom

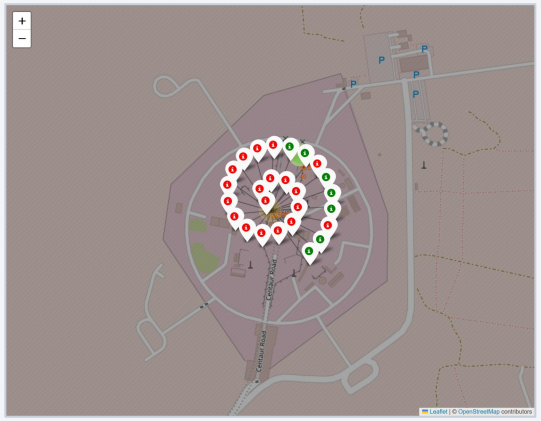
- 3 sites in Florida (~28.5°N), 1 in California (34.6°N)
- None near the equator, but Florida is the southernmost US option, maximising Earth-rotation boost
- All sit on the coast, so launches fly over the ocean

KEY INSIGHT All sites are on the coast, and the Florida sites are as far south as the mainland US allows.

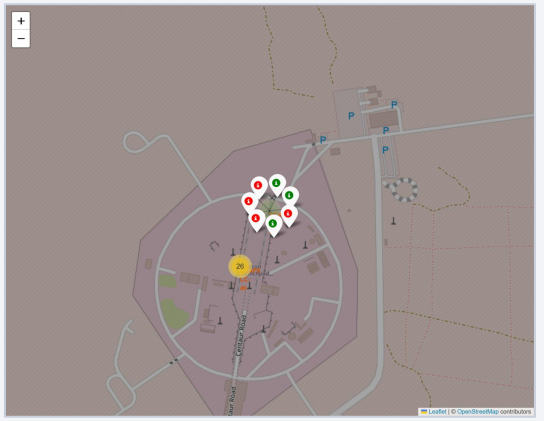
Colour-Labelled Launch Outcomes per Site



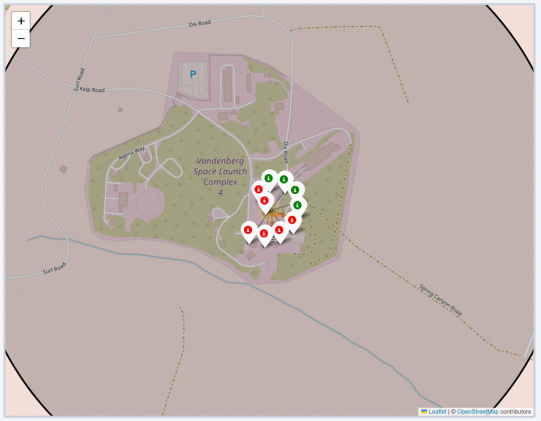
KSC LC-39A · 10 of 13 landed (76.9%)



CCAFS LC-40 · 7 of 26 (26.9%)



CCAFS SLC-40 · 3 of 7 (42.9%)

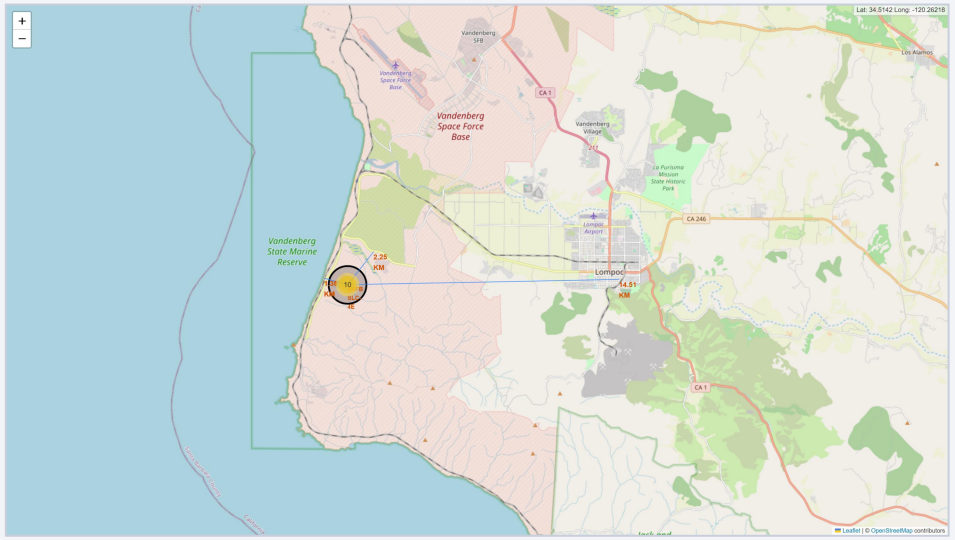


VAFB SLC-4E · 4 of 10 (40.0%)

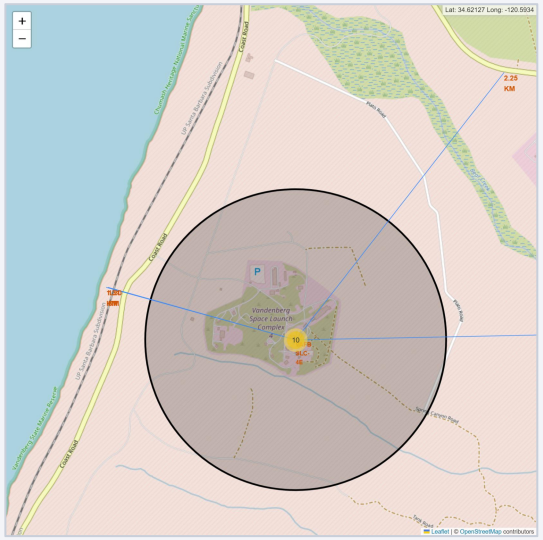
- MarkerCluster groups each site's launches; expanding a cluster shows one marker per launch: green = landed (class 1), red = failed (class 0)
- KSC LC-39A is overwhelmingly green, while CCAFS LC-40 (the busiest pad, with most early flights) is mostly red

KEY INSIGHT KSC LC-39A has the best record. CCAFS LC-40 has the lowest, partly because most early flights launched there.

VAFB SLC-4E Proximities and Distances



PolyLines to the city of Lompoc (14.51 km)

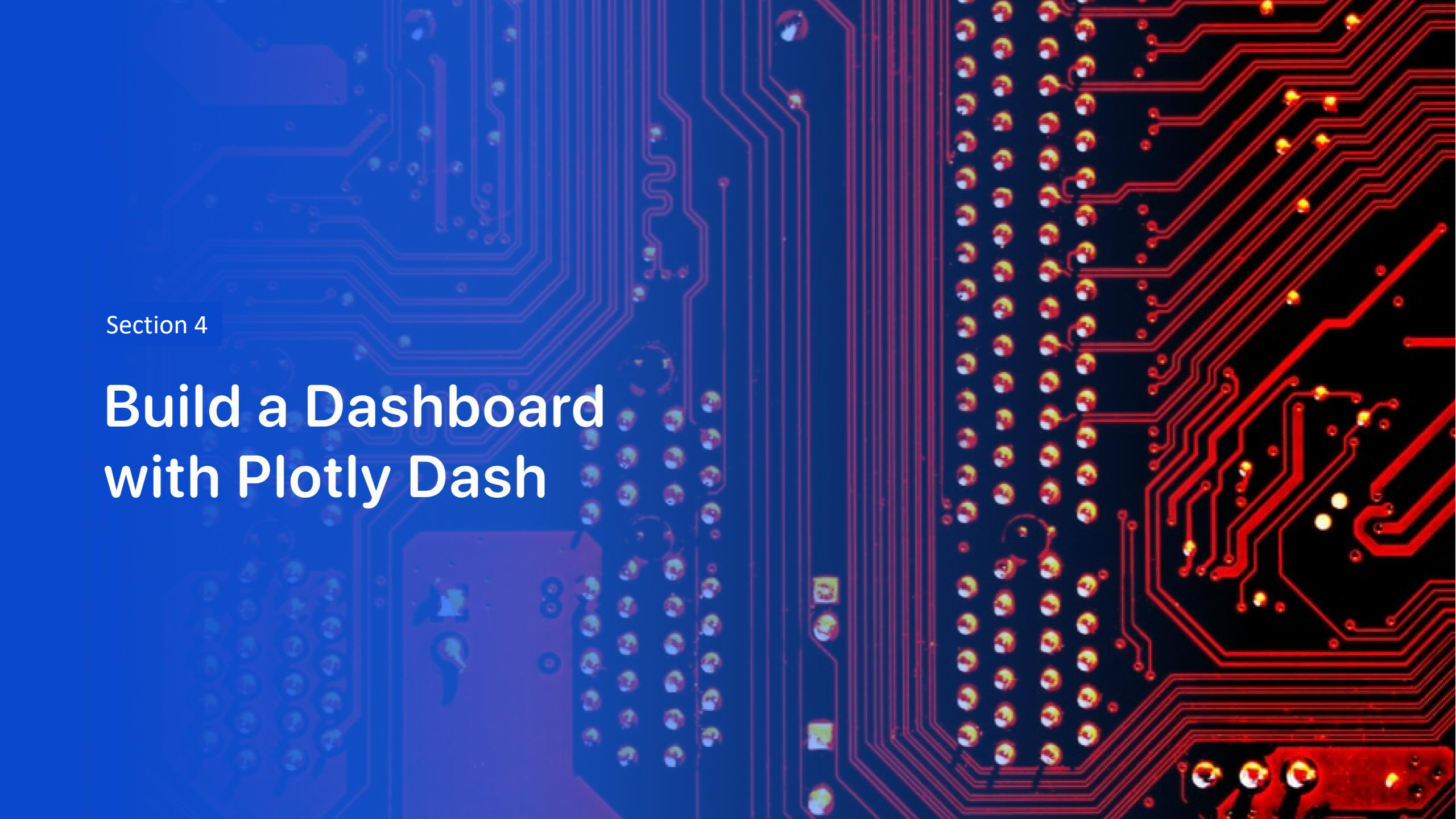


Coast, railway and highway

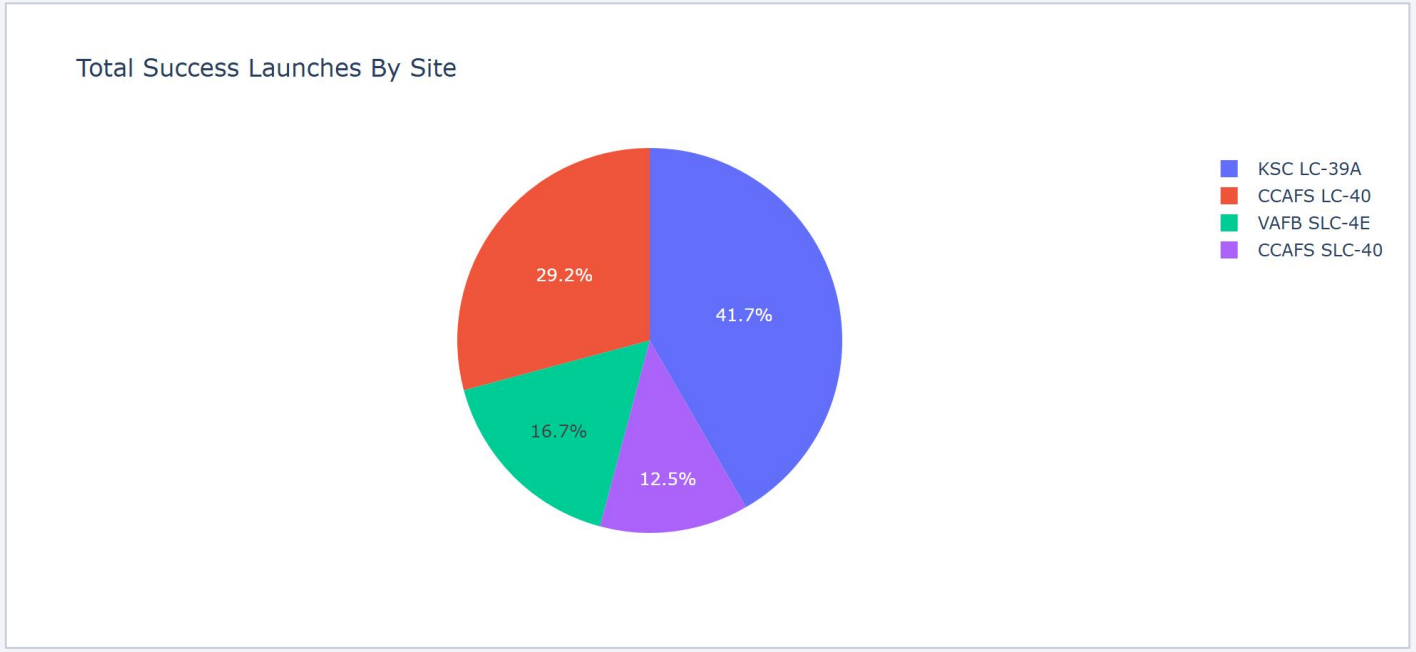
Proximity	Distance
Coast road	1.17 km
Railway	1.30 km
Coastline	1.31 km
Highway	2.25 km
Lompoc (city)	14.51 km

- Yes, close to railways and highways (1.3–2.3 km) for moving rockets, propellant and crews
- Yes, close to the coastline (~1.3 km) so ascent and any debris fall over the ocean; far from cities (~14.5 km) for public safety

KEY INSIGHT The launch site is close to the coast, railway and highway, but far from the nearest city.


Section 4
Build a Dashboard with Plotly Dash

Launch Success Count for All Sites



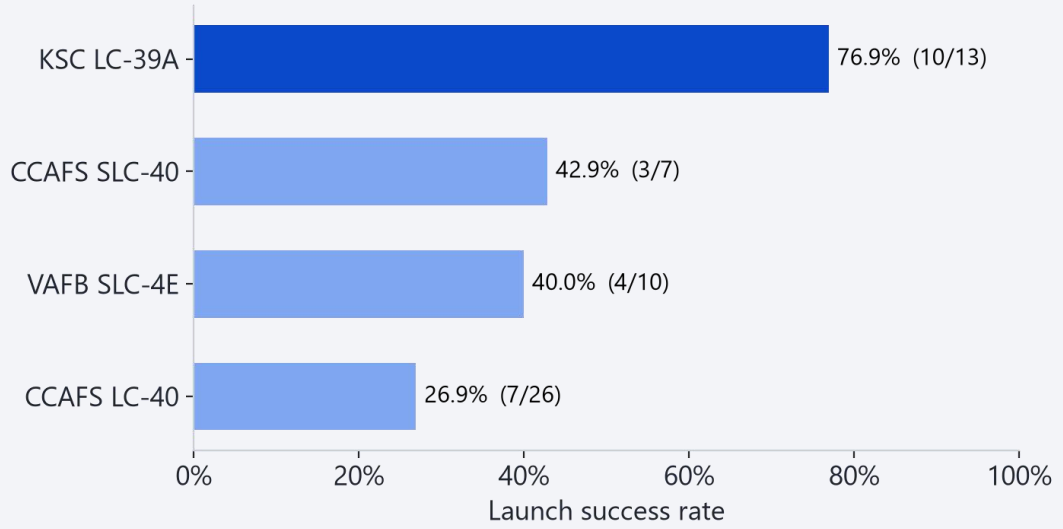
- **KSC LC-39A:** 41.7% of all successful launches (10)
- **CCAFS LC-40:** 29.2% (7), despite the most launches (26)
- **VAFB SLC-4E:** 16.7% (4)
- **CCAFS SLC-40:** 12.5% (3)
- Note: a large share is not the same as a high success rate. CCAFS LC-40 has many successes because it launched the most.

KEY INSIGHT KSC LC-39A has the most successful launches of all sites.

KSC LC-39A: Highest Launch Success Ratio



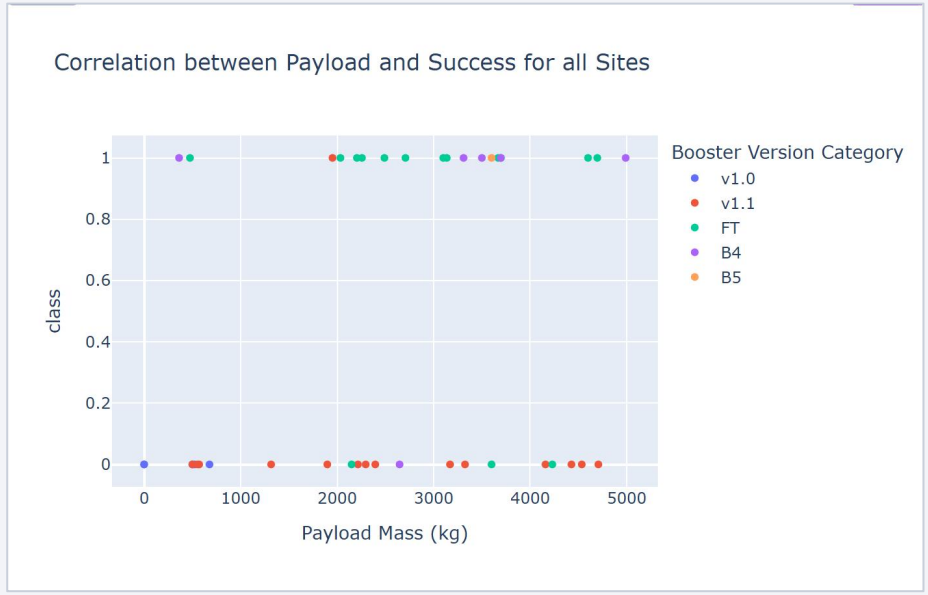
Dashboard with site = KSC LC-39A (1 = success, 0 = failure)



- 76.9% success (10 of 13), 23.1% failure (3)
- Next best: CCAFS SLC-40 at 42.9%, less than 4 in 7

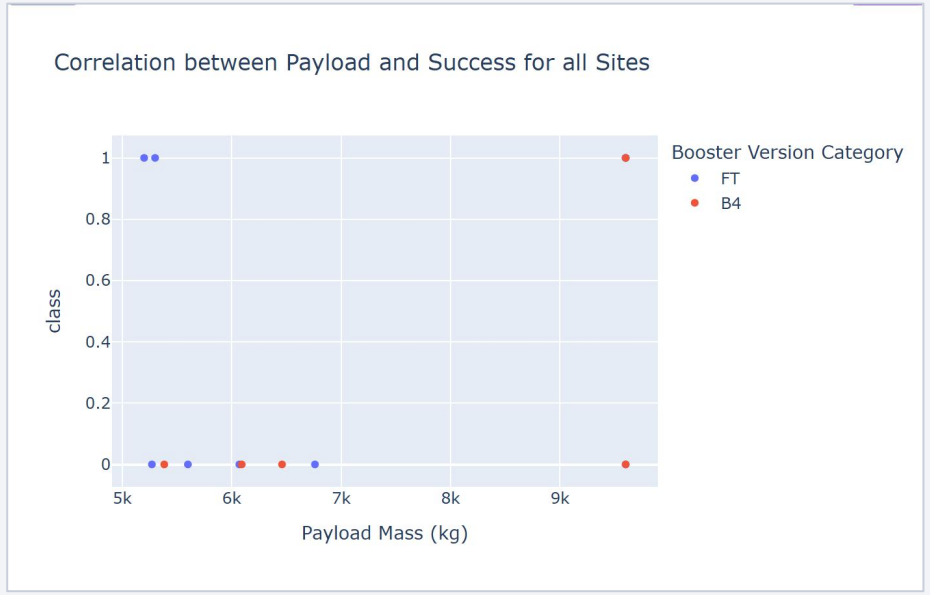
KEY INSIGHT KSC LC-39A's success rate (76.9%) is almost three times that of CCAFS LC-40 (26.9%).

Payload vs. Launch Outcome for All Sites



Slider 0–5,000 kg · 42 launches · 45.2% success

- **Best payload range:** 2,000–5,000 kg (55.2% success, 29 launches)
- **Heavier 5,000–10,000 kg:** 35.7%, fewer and harder missions

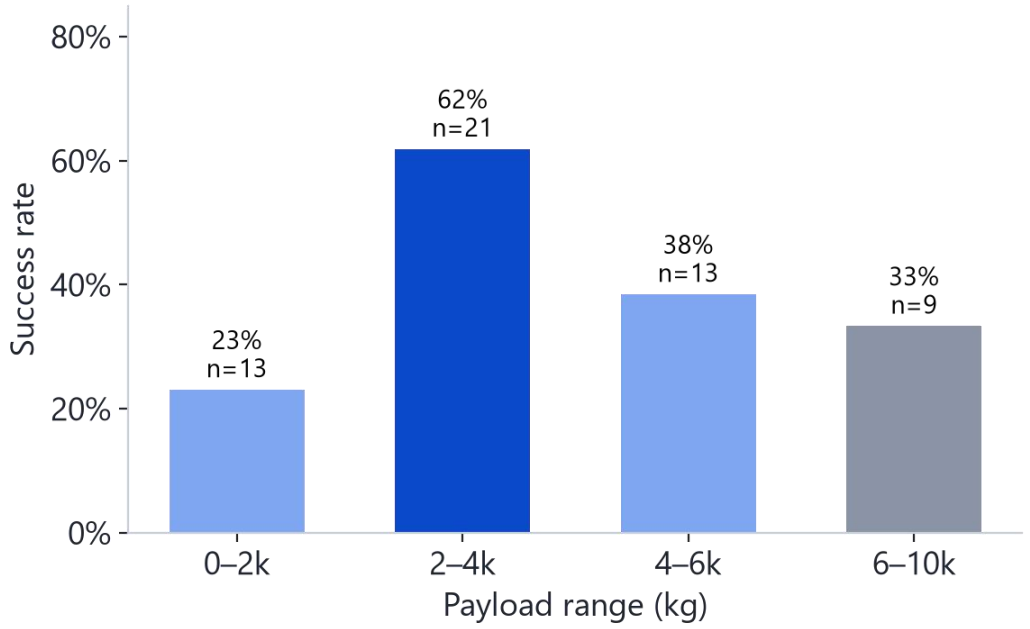


Slider 5,000–10,000 kg · 14 launches · 35.7% success

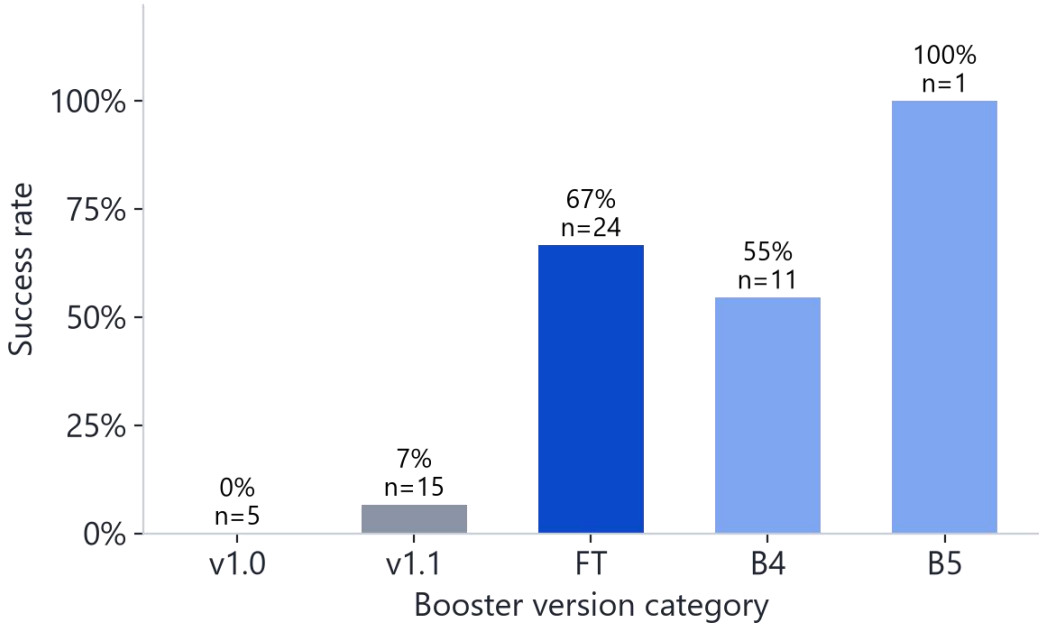
- **Best booster:** FT, 16 of 24 successful (66.7%)
- **Worst:** v1.0 (0 of 5) and v1.1 (1 of 15)

KEY INSIGHT Payloads between 2,000 and 5,000 kg and FT boosters have the highest success rates.

Dashboard Deep-Dive: Payload Range and Booster



Success rate by payload range (dashboard data, 56 launches)



Success rate by booster version category

Computed from spacex_launch_dash.csv to quantify what the dashboard scatter shows; B5 has a single launch in this dataset.

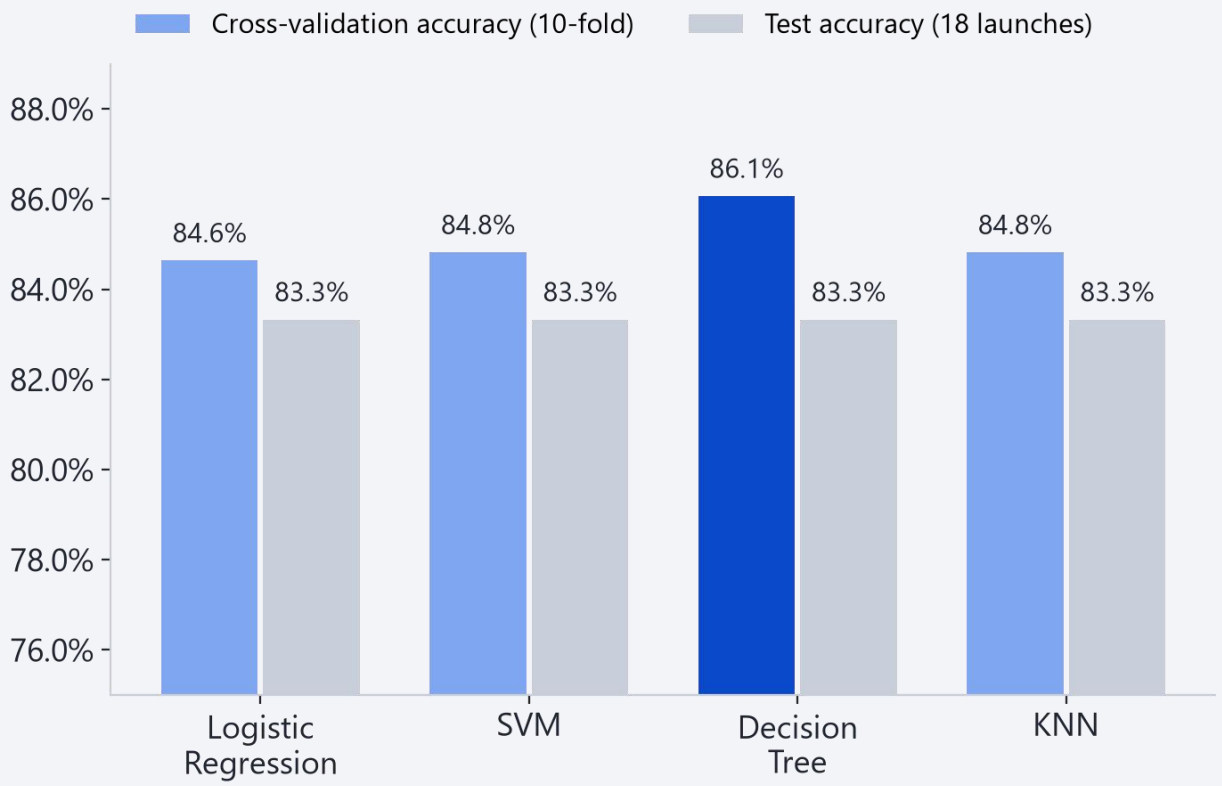
INSIGHT Booster version makes a bigger difference than payload. v1.0 and v1.1 almost never landed, while FT landed 67% of the time.

Section 5

Predictive Analysis (Classification)



Classification Accuracy

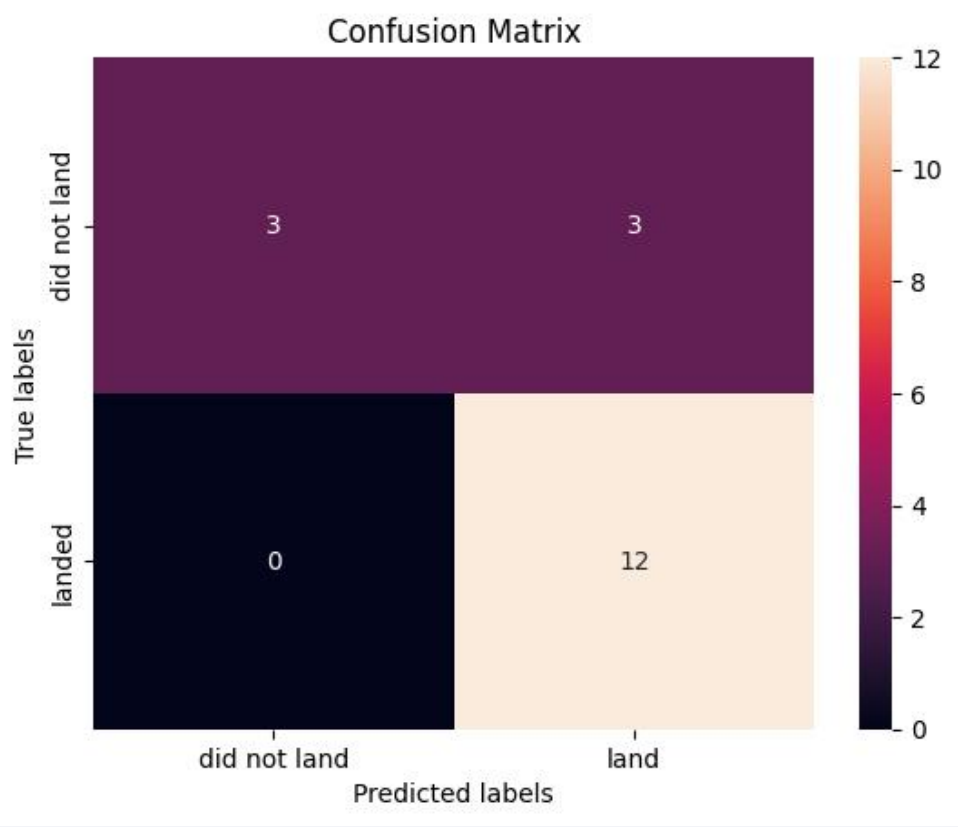


Model	CV acc.	Test acc.
Logistic Regression	84.6%	83.3%
SVM (sigmoid)	84.8%	83.3%
Decision Tree	86.1%	83.3%
KNN (k = 10)	84.8%	83.3%

- **Best: Decision Tree** with the highest cross-validation accuracy (86.1%)
- Params: gini, max_depth 4, sqrt features, min_samples_leaf 2, min_samples_split 5
- All four models tie on the 18-launch test set (15 of 18 correct)

KEY INSIGHT The Decision Tree has the best CV score, but on the small test set (18 launches) all four models score the same.

Confusion Matrix



12 **True positives**
landed, predicted landed

3 **True negatives**
failed, predicted failed

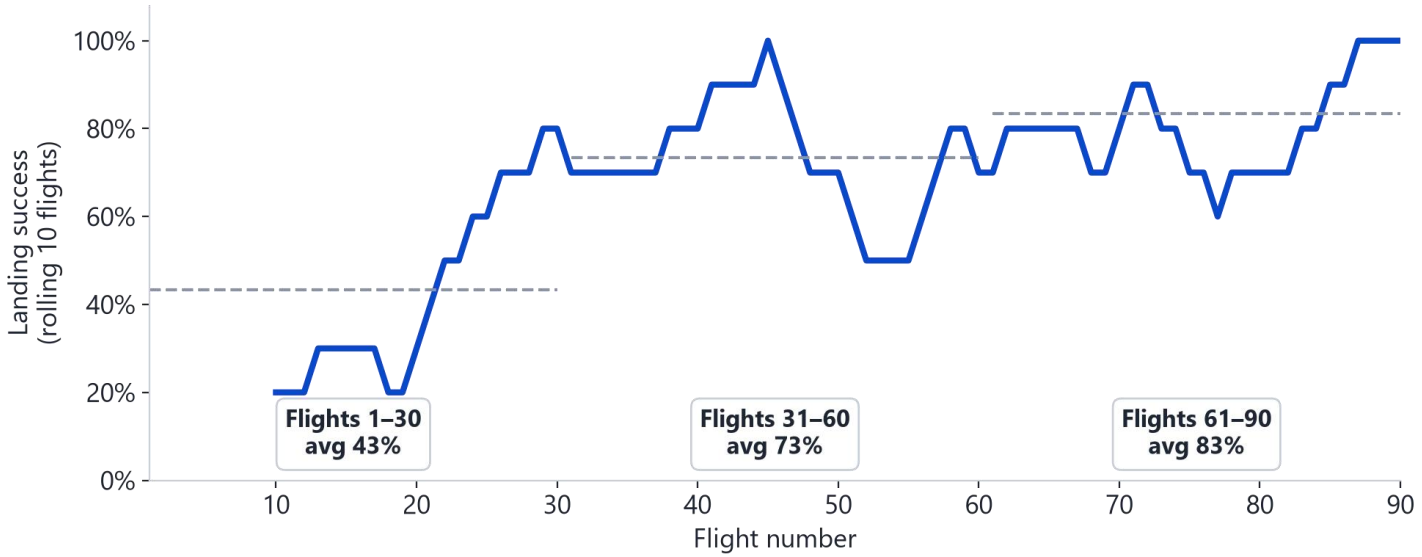
3 **False positives**
failed, predicted landed

0 **False negatives**
landed, predicted failed

- **Recall (landed) 100%,** precision 80%, specificity 50%
- The model never misses a real landing, but labels half of the actual failures as landings
- All four models produce this identical matrix

KEY INSIGHT All of the model's mistakes are false positives, so it is slightly too optimistic about landings.

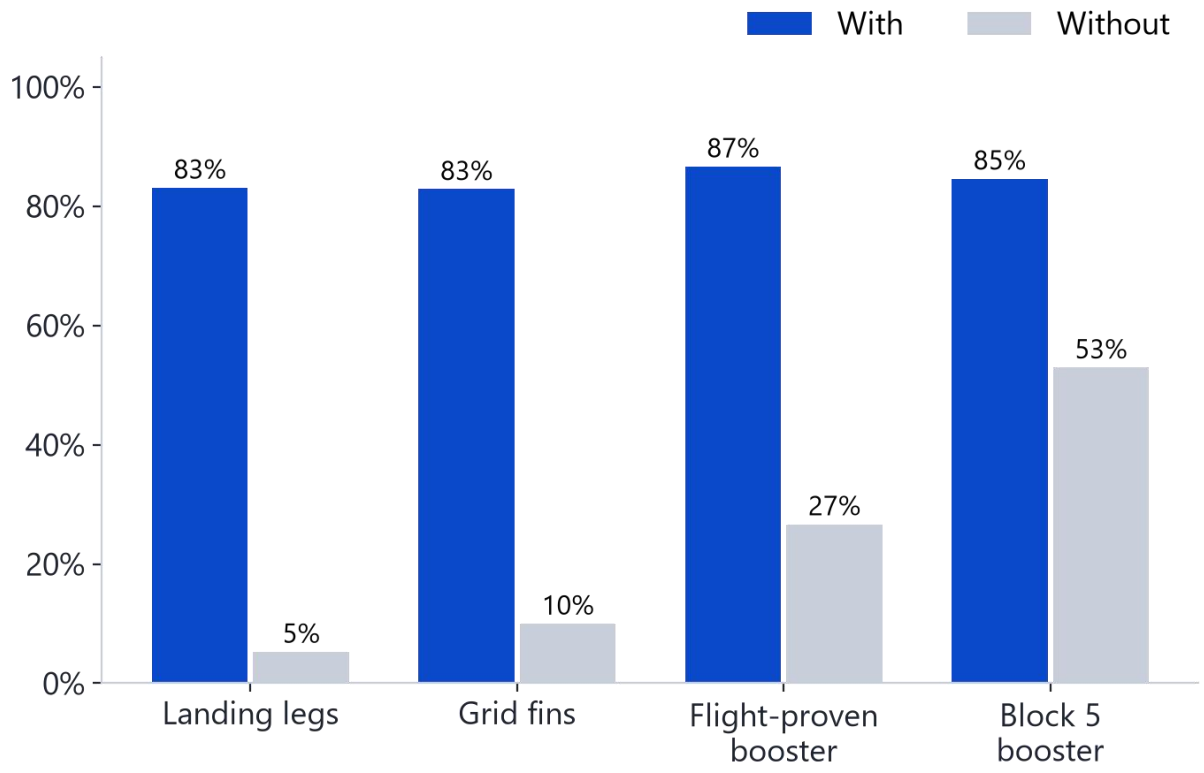
Innovative Insight 1: The Experience Curve



- **43%, 73% and 83%:** landing success for flights 1–30, 31–60 and 61–90 (chart shows a rolling 10-flight average)
- **Every site and orbit** improves with flight number, so some of the payload, site and orbit effects are really experience effects
- **What it means:** a model trained mostly on early flights will underestimate landings today, so recent launches should count more

INSIGHT Landing success went up with each group of 30 flights.

Innovative Insight 2: Hardware and Reuse Drive Landings



- **Landing legs:** 83% success with legs vs. 5% without, since a landing can only happen when one is attempted
- **Grid fins:** 83% vs. 10%, the same pattern as landing legs
- **Flight-proven boosters** land 87% of the time vs. 27% for first-flight cores
- **Block 5** boosters land 85% vs. 53% for earlier blocks
- Computed from dataset_part_2 (90 launches).

INSIGHT Re-used boosters landed more often than new ones.

Innovative Insight 3: What This Means for Space Y's Bids

$103M
price gap per launch ($165M vs $62M)

~83%
of recent flights land (61–90)

3 / 6
failures the model predicts as landings

Use the model to price bids

- Predict landing probability per mission (orbit, payload, booster block, site)
- If a mission is likely to land, expect SpaceX to price near $62M; if not, its price advantage is smaller
- Target GTO and other high-energy missions, where landing success is lowest (52%)

Make the model decision-ready

- 18 test launches is too few: use repeated stratified CV and add 2021+ launches
- Add features like booster block, reuse count and landing type, and weight recent flights
- Output calibrated probabilities, not just labels, to reduce costly false positives

INSIGHT A competitor has the best chance on missions where SpaceX is less likely to recover the booster.

Conclusions

- **Experience is the strongest driver:** landing success rose from 0% (2010–13) to 84–90% (2019–20); and from 43% to 83% between the first and last 30 flights
- **Site:** KSC LC-39A has the highest success rate (76.9%); CCAFS LC-40 the lowest (26.9%), reflecting the early years
- **Orbit & payload:** ES-L1, GEO, HEO, SSO 100% (small samples), GTO hardest at 52%; heavy payloads succeed on modern boosters
- **Booster:** v1.0/v1.1 rarely landed; FT and Block 5 made landings routine; flight-proven boosters land more often
- **Locations:** all sites are coastal, 1–2 km from transport links and ~15 km from cities
- **Model:** tuned Decision Tree selected (86.1% CV, 83.3% test); no missed landings, 3 false positives, so estimates are optimistic

66.7%	83%	76.9%	86.1%
overall landing success (90 launches)	landing success, flights 61–90	KSC LC-39A launch success	best CV accuracy (Decision Tree)

KEY INSIGHT Public launch data is enough to predict landings reasonably well and to support Space Y's pricing.

Appendix

REPOSITORY AND NOTEBOOKS

- **Repository:** SolexJ1000/SpaceX-Falcon-9-first-stage-Landing-Prediction
- **Data collection: SpaceX API:** 01_data_collection_api.ipynb
- **Data collection: web scraping:** 02_data_collection_webscraping.ipynb
- **Data wrangling:** 03_data_wrangling.ipynb
- **EDA with visualization:** 05_eda_visualization.ipynb
- **EDA with SQL:** 04_eda_sql.ipynb
- **Interactive map: Folium:** 06_interactive_map_folium.ipynb
- **Dashboard: Plotly Dash:** spacex_dash_app.py
- **Predictive analysis:** 07_machine_learning_prediction.ipynb

KEY CODE: LANDING LABEL AND MODEL TUNING

```
bad_outcomes = set(landing_outcomes.keys()[[1,3,5,6,7]])
landing_class = [0 if o in bad_outcomes else 1
                 for o in df["Outcome"]]
df["Class"] = landing_class


tree_cv = GridSearchCV(DecisionTreeClassifier(),
                       parameters, cv=10)
tree_cv.fit(X_train, Y_train)
tree_cv.score(X_test, Y_test)   # 0.833
```

- Datasets: dataset_part_1/2/3.csv, Spacex.csv, spacex_launch_geo.csv, spacex_launch_dash.csv
- Extra charts (insights, accuracy, site and booster rates) recomputed in Python from these lab datasets

Thank you!

